

Short answers reconstructed from two cropped photos of the same question paper.

Q1 Bloch functions under $\mathbf{R}$ and $\mathbf{G}$ translations

(a) By Bloch's theorem, $\psi_{n,\mathbf{K}}(\mathbf{r}) = e^{i\mathbf{K}\cdot\mathbf{r}}u_{n,\mathbf{K}}(\mathbf{r})$ with $u_{n,\mathbf{K}}(\mathbf{r} + \mathbf{R}) = u_{n,\mathbf{K}}(\mathbf{r})$ for any direct lattice vector $\mathbf{R}$. Hence

$$\psi_{n,\mathbf{K}}(\mathbf{r} + \mathbf{R}) = e^{i\mathbf{K}\cdot\mathbf{R}}\psi_{n,\mathbf{K}}(\mathbf{r}).$$

Since $\mathbf{G}$ is a reciprocal lattice vector, $e^{i\mathbf{G}\cdot\mathbf{r}}$ has the periodicity of the direct lattice, so it can be absorbed into the periodic part: $u_{n,\mathbf{K}+\mathbf{G}}(\mathbf{r}) = e^{-i\mathbf{G}\cdot\mathbf{r}}u_{n,\mathbf{K}}(\mathbf{r})$, giving

$$\psi_{n,\mathbf{K}+\mathbf{G}}(\mathbf{r}) = \psi_{n,\mathbf{K}}(\mathbf{r}) \quad \text{for all } \mathbf{r}.$$

So $\mathbf{K}$ and $\mathbf{K} + \mathbf{G}$ label the *same* Bloch state; combined with the result above,

$$\psi_{n,\mathbf{K}+\mathbf{G}}(\mathbf{r} + \mathbf{R}) = e^{i\mathbf{K}\cdot\mathbf{R}}\psi_{n,\mathbf{K}}(\mathbf{r}).$$

(b) Because $\psi_{n,\mathbf{K}+\mathbf{G}}$ and $\psi_{n,\mathbf{K}}$ are identical states, their eigenvalues coincide:

$$\varepsilon_n(\mathbf{K} + \mathbf{G}) = \varepsilon_n(\mathbf{K}),$$

i.e. the band energy is periodic in reciprocal space with the periodicity of the reciprocal lattice ($\mathbf{K}$ need only be specified within the first Brillouin zone).

Q2 Intrinsic Fermi level in a 1-D semiconductor

For a d -dimensional semiconductor the conduction/valence effective density of states scales as $N_{c,v} \propto m^{d/2}$, so setting $n = p$ gives

$$E_F - E_{\text{mid-gap}} = \frac{k_B T}{2} \ln \frac{N_v}{N_c} = \frac{d k_B T}{4} \ln \frac{m_h}{m_e}.$$

For $d = 1$:

$$\Delta E = \frac{k_B T}{4} \ln \left(\frac{m_h}{m_e} \right).$$

With $m_h/m_e = 1.25$ and room temperature ($T = 300 \text{ K}$, $k_B T = 8.617 \times 10^{-5} \text{ eV/K} \times 300 \text{ K} = 0.02585 \text{ eV}$):

$$\Delta E = \frac{0.02585}{4} \ln(1.25) = \frac{0.02585}{4} (0.2231) \approx 1.44 \times 10^{-3} \text{ eV} \approx 1.44 \text{ meV}.$$

Since $m_h > m_e$, $\Delta E > 0$: the intrinsic Fermi level sits about **1.44 meV above mid-gap**, shifted slightly toward the conduction band.

Q3 3 valence electrons/atom: simple cubic vs. FCC

Simple cubic: A monatomic simple-cubic crystal has exactly **1 atom per primitive cell**. Each band can hold 2 electrons (spin) per primitive cell, so an even number of valence electrons/cell is needed to exactly fill an integer number of bands and leave a gap at E_F . With 3 valence electrons/atom (odd), the lowest band is completely filled (2 electrons) and the next band is exactly **half-filled** (1 electron) — there is necessarily a partially occupied band, so E_F lies inside a band. This is a metal, not a semiconductor.

FCC: The FCC Bravais lattice is also monatomic with **1 atom per primitive cell** (the 4-atom count refers to the conventional cubic cell, not the primitive cell). The same electron-counting

parity argument applies: 3 electrons/primitive cell is still odd, so a band must be left partially filled. Changing the lattice from simple cubic to FCC does not change this parity argument, so the element **remains metallic in FCC as well** (e.g. real trivalent FCC metals like Al are indeed conductors, not semiconductors). A gap at E_F for an elemental (monatomic-basis) crystal is only possible when the number of valence electrons per primitive cell is even.

Q4 [Hard]

Only fragments of this question survive across the two crops:

“The intrinsic carrier concentration of [an] FCC semiconductor at room temperature ... indirect band gap of $E_g = 0.66 \text{ eV}$ and relative dielectric constant $\kappa = \dots$, $n_i \approx 3 \times 10^{13} \text{ cm}^{-3}$. Effective mass measurement in a magnetic field of $B = 100 \dots$ ”

The actual question being asked (and its mark weight) is cut off in both photos, so it is left unsolved here rather than guessed at.